Assignment Title: Live Polling System using Angular Signals

Objective

Build a real-time polling application where users can vote on options and instantly see updated results using Angular Signals.

Problem Statement

Develop a Live Polling System where:

- A question is displayed with multiple options
- Users can vote for one option
- Vote counts update immediately without page refresh
- Results are dynamically recalculated using Angular Signals

Requirements

1. Poll Display

- Display a poll question (e.g., "Which technology do you prefer?")
- Show at least 4 options
- Each option should have a vote button

2. Voting Functionality

- When a user clicks a vote button:
    - The vote count for that option should increase
    - The UI should update instantly

3. Live Results (Using Signals)

- Maintain vote counts using signal()
- Compute total votes using computed()
- Calculate percentage of votes for each option dynamically
- Display results like:
    - Option A – 40% (20 votes)

4. Prevent Multiple Votes (Basic Logic)

- Once a user votes, disable further voting (simple boolean flag is enough)

5. Reset Poll

- Add a “Reset Poll” button
- Reset all votes to zero
- Re-enable voting

Technical Constraints

- Use Angular (latest version)
- Use Angular Signals:
    - signal() for state
    - computed() for derived values
    - effect() (optional) for logging or side effects
- Do not use RxJS for state management in this assignment
- Follow component-based architecture

Suggested Component Structure

- Poll
    - Displays question and options
    - Handles voting logic
    - Displays live results

Example Data

pollQuestion = "Which frontend framework do you prefer?";

options = [
 { id: 1, name: "Angular", votes: 0 },
 { id: 2, name: "React", votes: 0 },
 { id: 3, name: "Vue", votes: 0 },

```
  { id: 4, name: "Svelte", votes: 0 }
];
```

Expected Output

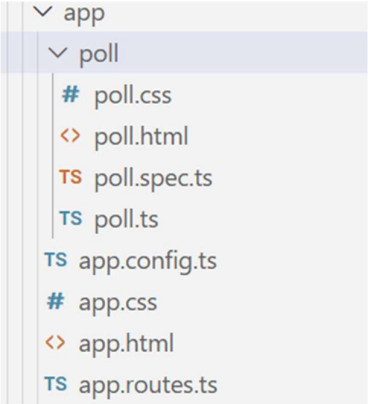
- User clicks on an option
- Vote count increases instantly
- Percentages update automatically
- Voting buttons become disabled after selection
- Reset button clears all data

Evaluation Criteria

Criteria	Marks
Correct use of signal()	**5**
Correct use of computed()	**5**
UI updates instantly without refresh	**5**
Voting logic and restrictions	**3**
Reset functionality	**2**
Total	**20**

Note: A Project structure would be given. You have to Add “Poll” component only. The name of the component is strict and if the name changes, then test might not evaluate yours

Folder Structure:



Generated poll class should be

export class PollComponent {

}

Deliverables: Simply app folder and inside app folder Poll folder. Other files of app like app.component.html, css, ts, spec files are not required.

You can discard sample1@matrimony.com as it will be handled by the evaluator while evaluating

